

shown to be more accurate than traditional ML learners. It is also important to note that extensive feature engineering, which has been associated with traditional ML learners, need not be done in the case of neural networks. Only core features are fed as input to the neural network and the network itself can learn additional features on its own. We will discuss the differences between traditional ML based learners and neural networks in our next column.

While we have so far been talking about inputs to individual neurons in a neural network, we can, in fact, treat the inputs coming into each layer of a neural network as a vector, thus simplifying the mathematical calculations. From now onwards, in our discussions, we will use the vector representation as inputs to a neural network. The weights of the connections between the different layers will be represented using matrices. A fully connected layer is in fact an implementation of a vector-matrix multiplication, where the incoming input vector X is multiplied by the weight matrix W of the connections.

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The search for an explanation led to a kernel bug that people have known about for quite some time, but haven't been able to resolve yet. However, the kernel option 'intel_idle.max_cstate=1' seemed to work most of the time. It reduced the frequency of the freezes.

A Phoronix article, [goo.gl/GcCiMe](http://www.phoronix.com/scan.php?page=item&category=linux&item=linux-kernel-bug), describes the problems many Linux users have faced with Bay Trail devices. Booting and getting the login prompt takes about 30 seconds. Logging into GNOME 3 is slow (15 to 30 seconds) and into KDE 5 is far worse. However, LXDE is very responsive.

Disabling the 'Enable compositor on startup' in the 'System Settings', 'Display and Monitor' and 'Compositor' settings makes KDE5 usable, though it does take a while (about a minute) to log in to the KDE desktop.

Wi-Fi

The Wi-Fi chip was not detected at first. A driver is available at <https://github.com/hadess/rtl8723bs>.

Since the development tools are not installed on the netbook, it was easier to build the driver (using the 'make' command) on the desktop, then copy the files on the netbook and run the command 'sudo make install'.

The Wi-Fi worked well. However, if the connection was unstable, the system's performance deteriorated sharply. This device is supposed to work with both Wi-Fi and Bluetooth, though the latter does not work at present.

Sound

The only audio device visible was 'Dummy'. The following code resulted in the sound card being visible to PulseAudio:

Download <https://github.com/plbossart/UCM/archive/t180taf-test>.

We will describe the mathematical notations for the neural networks in the next column.

If you have any favourite programming questions/software topics that you would like to discuss on this forum, please send them to me, along with your solutions and feedback, at sandyasm_AT_yahoo_DOT_com. Till we meet again next month, happy programming! 

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zip

To enable UCM(Use Case Manager) on RT5640 platforms, copy directory to UCM location:

```
$ sudo cp -rf bytcr-rt5640 /usr/share/alsa/ucm
```

Though several solutions were suggested, none worked. It was a surprise to discover that people have been actively trying to solve the issues related to the kernel lock and sound issues since 2014, and there's still no solution.

The USB devices I used, e.g., a 3G dongle, worked very well. The system was very nice to use when it was behaving well!

However, sound remained an issue. Even Bluetooth speakers could not be used.

On the other hand, given the high level of activity in the mailing lists, the chance of some solutions being worked out in the near future are high. However, if I were to buy a system today, I would stay away from one with the Bay Trail CPU in it. 

References

- [1] No sound: https://bugzilla.kernel.org/show_bug.cgi?id=86581
- [2] Crashes: https://bugzilla.kernel.org/show_bug.cgi?id=109051

By: Dr Anil Seth

The author has earned the right to do what interests him. You can find him online at <http://sethanil.com> and <http://sethanil.blogspot.com>, or you can reach him via email at anil@sethanil.com.